

AFRICAN INSTITUTE FOR MATHEMATICAL SCIENCES

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Assignment 1

Course: *Numerical Linear Algebra (NLA)*

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Question 1

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$. Show that the following conditions are equivalent

1. $\mathbf{A}$ has an inverse $\mathbf{A}^{-1}$,
2. $\text{rank}(\mathbf{A}) = n$,
3. $\text{range}(\mathbf{A}) = \mathbb{R}^n$,
4. $\text{null}(\mathbf{A}) = \{\mathbf{0}\}$,
5. 0 is not an eigenvalue of $\mathbf{A}$,
6. 0 is not a singular of $\mathbf{A}$,
7. $\det(\mathbf{A}) \neq 0$.

Question 2

Show that if a matrix $\mathbf{A}$ is both triangular and unitary, then it is diagonal.

Question 3

Let $S \in \mathbb{C}^{n \times n}$ be skew-hermitian, i.e. $S^* = -S$.

1. Show by using the previous exercise that the eigenvalues of S are pure imaginary.
2. Show that $\mathbf{I} - \mathbf{S}$ is nonsingular.
3. Show that the matrix $\mathbf{Q} = (\mathbf{I} - \mathbf{S})^{-1}(\mathbf{I} + \mathbf{S})$, known as the Caley transform of S , is unitary.

Question 4

If $\mathbf{u}$, and $\mathbf{v}$ are m -vectors, the matrix $\mathbf{A} = \mathbf{I} + \mathbf{u}\mathbf{v}^*$ is known as a rank-one perturbation of the identity. Show that if $\mathbf{A}$ is nonsingular, then its inverse has the form $\mathbf{A}^{-1} = \mathbf{I} - \alpha\mathbf{u}\mathbf{u}^*$ for some scalar α , and give an expression for α . For what $\mathbf{u}$ and $\mathbf{v}$ is $\mathbf{A}$ singular? If it is singular, what is the $\text{null}(\mathbf{A})$?

Question 5

Write a Julia program that performs matrix-vector multiplication two different ways: (a) using the built-in Julia command $\mathbf{b} = \mathbf{A} \star \mathbf{x}$, and (b) using loops, as follows.

```
b = zeros(n) # Create a vector of zeros with length n
for j in 1:n
    for i in 1:n
        b[i] = b[i] + A[i, j] * x[j] # Update b[i] by adding A[i, j] * x[j]
    end
end
```

Perform three experiments using different vector and matrix sizes (3x3), (10x10), and (100x100). Time (by using the julia @time function) the two different methods on matrices of various sizes. Which method is faster?

Question 6

Find by hand the singular value decomposition (SVD) of the matrix and check your result using Julia.

$$\mathbf{B} = \frac{1}{15} \begin{pmatrix} 14 & 2 \\ 4 & 22 \\ 16 & 13 \end{pmatrix}$$